# CEO Sentiment and Strategic Stance of Foreign Firms Toward China

*A Time-Series Analysis of Corporate Risk Posturing, 2015–2025*


By

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*Generative AI was used for batch application of the coding scheme, Python and LaTeX code generation. All source retrieval, final stance classifications, analytical interpretations, and written conclusions reflect the author's independent judgement.

# Executive Summary

| 175 | 9 | 3 | 11 | 0.931 |
|---|---|---|---|---|
| Executive Statements Coded | Multinational Firms | Industry Sectors | Years of Coverage | Intercoder Reliability ($\alpha$) |

This report constructs a longitudinal dataset measuring CEO-level strategic stance toward China across nine multinational firms — spanning healthcare, automotive, and information technology, over the period 2015 to 2025. Executive commentary from earnings calls, investor presentations, and annual reports was classified on a five-point ordinal scale ranging from *Committed Expansion* (1) to *Strategic Retreat* (5), using an LLM-assisted coding methodology validated by intercoder reliability of $\alpha = 0.931$.

## Central Finding

Strategic posture toward China varies systematically by **policy exposure type** rather than by sector alone. Firms subject to US export controls (NVIDIA) undergo forced, reactive sentiment shifts. Firms facing host-country regulatory dynamics such as pharmaceutical pricing reforms (Pfizer, AstraZeneca, Novartis) or domestic competitive pressure (Toyota, BMW, Tesla) — demonstrate more stable, anticipatory positioning.

## Three Patterns That Define the Decade

**1. Technology sector polarisation.** NVIDIA's stance moved from *Optimistic Engagement* (Score 2, 2015–2022) to *Active Hedging / Strategic Retreat* (Score 4–5, 2023–2025), driven entirely by successive rounds of US semiconductor export controls culminating in a \$4.5 billion inventory write-down. Meanwhile, Microsoft and SAP maintained moderate positions through compliant cloud architectures. The within-sector variance in Technology (SD = 0.73) is the highest across all three sectors. *Export controls create winners and losers within the technology sector, not across it uniformly.*

**2. Healthcare resilience and counter-cyclical investment.** All three pharmaceutical firms maintained mean scores below 2.0 throughout the decade. There is no statistically significant pre- vs post-2022 shift ($p = 0.878$). Novartis announced a \$100 million radioligand therapy facility; AstraZeneca opened its 6th R&D centre in Beijing in 2025; Pfizer ranked China as its #1 international market. Volume-based procurement creates margin pressure but not strategic retreat. *The pharmaceutical sector operates in a structurally different geopolitical reality than semiconductors.*

**3. Automotive adaptation without retreat.** The Automotive sector shows a significant upward shift post-2022 ($p = 0.009$, $d = 1.16$), but critically, this reflects *localisation*, not withdrawal. BMW acquired a 75% majority stake in its Chinese joint venture (the first foreign OEM to do so); Toyota acknowledged "we really endured" but maintained profitability on par with domestic competitors; Tesla's Shanghai Gigafactory remains its most cost-efficient plant globally. *"Local for local" rhetoric serves dual functions—operational efficiency and geopolitical risk management.*

## Sector Comparison At a Glance

| Sector | Pre-2022 | Post-2022 | Δ | Significant? | Primary Driver |
|---|---|---|---|---|---|
| Healthcare | 1.96 | 1.93 | −0.03 | No ($p$=0.878) | VBP pricing reforms |
| Automotive | 1.65 | 2.23 | +0.58 | Yes ($p$=0.009) | BEV competition |
| Technology | 2.27 | 3.11 | +0.84 | Yes ($p$=0.021) | Export controls |

## Implications

For **corporate strategists**, these findings suggest that China risk assessment must be disaggregated by regulatory exposure type, not by industry label. For **policymakers**, the evidence shows that export controls effectively constrain targeted firms but do not produce uniform sectoral withdrawal. For **researchers**, the dataset demonstrates that LLM-assisted content analysis can achieve publication-quality intercoder reliability for complex strategic stance classification, opening a scalable pathway for longitudinal studies of executive communications.

Perhaps the most striking finding is what *did not* happen: despite a decade of escalating US-China tension, the overall mean stance score across all nine firms remained at 2.09, placing it firmly in *Optimistic Engagement* territory. The narrative of broad multinational retreat from China is not supported by the executive communications data. What the data reveal instead is a highly selective, policy-specific recalibration, with true strategic retreat concentrated almost entirely in targeted technology firms like NVIDIA facing specific regulatory actions such as semiconductor export controls.

# Contents

# 1 Introduction

Over the past decade, the strategic environment for foreign firms operating in China has undergone profound transformation. The US-China trade war (2018), cybersecurity and data sovereignty legislation (2017–2021), semiconductor export controls (2022–2025), and intensifying domestic competition in electric vehicles and pharmaceuticals have collectively reshaped how multinational executives frame their China strategies in public communications.[1]

This report constructs a time-series representation of CEO-level strategic stance toward China, enabling an assessment of how executive ideology and strategic framing changed over this period. The analysis covers nine firms across three sectors: Healthcare (Pfizer, AstraZeneca, Novartis), Automotive (Toyota, BMW, Tesla), and Information Technology (Microsoft, SAP, NVIDIA)—drawing exclusively on public, English-language executive commentary.

The contribution is threefold. First, the construction of a structured, reproducible dataset with 175 coded observations spanning 80 firm-year observations. Second, the application and validation of LLM-Assisted Content Analysis (LACA) for strategic stance measurement, achieving intercoder reliability of $\alpha = 0.931$. Third, the identification of systematic cross-sector patterns in executive China positioning that are driven by policy exposure type rather than sector membership alone.

# 2 Methodology

## 2.1 Theoretical Foundation

The measurement premise rests on **Upper Echelons Theory** (Hambrick & Mason, 1984), which posits that organisational outcomes, including strategic decisions about market commitment, geographic expansion, and risk management reflect the cognitive frames and strategic orientations of top management teams. CEO and C-suite communications on earnings calls serve as observable indicators of strategic intent, as executives must articulate and justify their firms' positioning to investors, analysts, and regulators.[2]

This theoretical grounding is consistent with a growing body of research using CEO discourse to measure strategic orientation. Huang et al. (2023) demonstrate that financial text contains extractable signals of managerial sentiment, while **?** apply similar techniques specifically to earnings call transcripts.

[1] The period 2015–2025 captures the full arc from the announcement of Made in China 2025 through the most recent rounds of US semiconductor export controls, providing a natural time window for longitudinal analysis.

[2] Earnings call transcripts are particularly valuable because the Q&A format creates adversarial pressure that limits purely aspirational messaging—analysts ask pointed questions about China-specific risks and revenue exposure.

## 2.2 Measurement Construction

### 2.2.1 The Five-Point Strategic Stance Scale

Executive commentary was classified on a deductive, five-point ordinal scale designed to capture the spectrum of strategic orientations a multinational CEO might adopt toward a major emerging market under geopolitical pressure.[3]

**Table 1:** Strategic Stance Classification Scheme

| Score | Label | Operational Definition |
|---|---|---|
| 1 | Committed Expansion | Executive signals active, unambiguous commitment to growing China operations. Language includes investment announcements, production expansion, and framing China as a core strategic market. |
| 2 | Optimistic Engagement | Executive maintains positive orientation without explicit expansion commitment. Language includes continued presence affirmation, opportunity framing, and constructive engagement tone. |
| 3 | Cautious Localisation | Executive acknowledges complexity but frames response as operational adaptation. Language includes local-for-local sourcing, compliance-focused investment, and selective participation. |
| 4 | Active Hedging | Executive explicitly signals diversification away from China exposure. Language includes de-risking, supply chain diversification, and contingency planning. |
| 5 | Strategic Retreat | Executive signals exit, divestment, or material withdrawal from China operations. |

### 2.2.2 Supplementary Coding Dimensions

Each coded observation received two additional classifications.[4]

**Confidence flag** (High/Medium): Indicates the degree of classification certainty. High-confidence observations feature unambiguous signal language where independent coders would converge; Medium-confidence observations contain mixed signals or context-dependent language.

**Driver type** (Strategic/Reactive/Ambiguous): Distinguishes proactive risk management from compliance-driven responses. A classification is Reactive when the executive explicitly references a triggering policy event (e.g., "the new export control regulations"); Strategic when positioning reflects anticipatory choice; and Ambiguous when the driver cannot be determined from the statement alone.

[3] The scale was developed prior to data collection based on the international business literature on foreign market commitment. The deductive (theory-driven) approach follows the codebook development methodology recommended by Krippendorff (2004).

[4] These supplementary dimensions enable richer analytical decomposition—for instance, testing whether reactive (policy-forced) shifts are larger in magnitude than strategic (anticipatory) shifts.

### 2.2.3 Aggregation to Firm-Year Level

Where multiple executive statements exist for a given firm-year (e.g., quarterly earnings calls), the firm-year stance score is computed as the arithmetic mean of individual observation scores for that year.[5] This produces a panel of 80 firm-year observations (9 firms $\times$ up to 11 years), enabling time-series visualisation and statistical trend testing.

## 2.3 Classification Procedure: LLM-Assisted Content Analysis

The classification followed the **LLM-Assisted Content Analysis (LACA)** framework (**?**), a methodology that leverages large language models as systematic coding instruments while preserving human interpretive authority. LACA has been validated against multiple benchmarks: Gilardi et al. (2023) demonstrate that LLMs outperform crowd workers for text annotation tasks, achieving higher accuracy and consistency across classification categories. Gale & Nicolajsen (2025) formalise a seven-step LACA workflow incorporating iterative prompt refinement and reliability thresholds of $\alpha > 0.80$, while Törnberg (2025) advocate a hybrid human-in-the-loop approach that combines LLM scalability with human interpretive depth.[6]

The procedure consisted of four steps:

1. **Codebook development:** The five-point scale (Table 1) was developed by the human researcher prior to data collection, grounded in upper echelons theory and international business literature.

2. **LLM classification:** Claude (Anthropic) applied the codebook to all 175 statements in structured batches of 5–8 statements, receiving the full codebook, sector-specific context, and policy timeline as input. For each statement, the LLM produced a stance score, confidence flag, driver type, and one-sentence justification.

3. **Human review:** The researcher independently reviewed every LLM classification, overriding where contextual judgement differed and documenting reasoning in coder notes.

4. **Reliability assessment:** Krippendorff's alpha for ordinal data was computed between the LLM and human classification vectors.

[5] The arithmetic mean is appropriate for ordinal data when the scale is treated as quasi-interval, which is standard practice in content analysis with five or more categories (Krippendorff, 2004). Sensitivity analysis using median scores produces substantively identical results.

[6] The choice of LACA over traditional manual content analysis or fully automated NLP approaches reflects a deliberate methodological trade-off: manual coding of 175 statements by multiple trained coders would be prohibitively time-intensive for a single-researcher project, while fully automated sentiment analysis (e.g., dictionary-based or FinBERT) lacks the contextual sensitivity required to distinguish, for instance, VBP-driven margin pressure from geopolitical risk hedging.

### 2.4 Intercoder Reliability

**Table 2:** Intercoder Reliability: LLM vs Human Classifications

| **Metric** | **Value** |
|---|---|
| Krippendorff's $\alpha$ (ordinal) | 0.931 |
| Krippendorff's $\alpha$ (nominal) | 0.867 |
| Exact agreement | 161/175 (92.0%) |
| Maximum disagreement | $\pm 1$ point |
| *By confidence flag* | |
| High confidence ($n = 117$) | $\alpha = 0.935$ |
| Medium confidence ($n = 58$) | $\alpha = 0.893$ |
| *By sector* | |
| Healthcare ($n = 94$) | $\alpha = 0.916$ |
| Automotive ($n = 46$) | $\alpha = 0.922$ |
| Technology ($n = 35$) | $\alpha = 0.956$ |

*Notes*: Krippendorff's $\alpha \geq 0.800$ is the recommended threshold for drawing tentative conclusions from coded data (Krippendorff, 2004). All 14 disagreements between the LLM and human coder were within $\pm 1$ point on the ordinal scale.

The ordinal $\alpha$ of 0.931 substantially exceeds the 0.800 threshold and approaches the 0.90 level considered "very good" in the content analysis literature.[7]

## 3 Data

### 3.1 Sources and Collection

Executive statements were sourced primarily from earnings call transcripts accessed through S&P Capital IQ, supplemented by company investor relations pages for annual reports and investor day presentations.[8] Only direct executive quotes or faithful paraphrases (marked with [P]) were coded; analyst commentary and journalist interpretation were excluded.

[7] The 14 disagreements clustered in borderline cases. For instance, whether AstraZeneca's 2021 acknowledgement of VBP-driven revenue decline constituted Cautious Localisation (LLM) or Optimistic Engagement with acknowledged headwinds (human reviewer). These edge cases illustrate the inherent fuzziness at category boundaries that Krippendorff's alpha is designed to account for.

[8] Capital IQ provides speaker-labelled transcripts with consistent formatting, enabling systematic identification of CEO, CFO, and regional executive commentary. Within each transcript, the researcher searched for China-related keywords ("China," "Chinese," "Beijing,") and extracted the surrounding context.

## 3.2 Coverage Summary

**Table 3:** Dataset Coverage by Firm

| Firm | Sector | $n$ | Year Range | Mean Score | Key Executive(s) |
|---|---|---|---|---|---|
| AstraZeneca | Healthcare | 44 | 2015–2025 | 1.95 | Pascal Soriot (CEO) |
| Pfizer | Healthcare | 26 | 2015–2025 | 2.00 | Albert Bourla (CEO) |
| Novartis | Healthcare | 24 | 2015–2025 | 1.75 | Vas Narasimhan (CEO) |
| BMW | Automotive | 20 | 2015–2025 | 1.80 | Oliver Zipse (CEO) |
| Tesla | Automotive | 12 | 2015–2025 | 1.83 | Elon Musk (CEO) |
| Toyota | Automotive | 14 | 2015–2025 | 2.29 | Yoichi Miyazaki (CFO) |
| NVIDIA | Technology | 12 | 2015–2025 | 3.00 | Colette Kress (CFO) |
| Microsoft | Technology | 12 | 2015–2023 | 2.75 | Satya Nadella (CEO) |
| SAP | Technology | 11 | 2015–2025 | 2.09 | Christian Klein (CEO) |
| **Total** | | **175** | **2015–2025** | **2.09** | |

*Notes*: Mean score is computed from human-reviewed stance classifications on the 1–5 ordinal scale. All firms exceed the minimum threshold of four observations. Coverage is densest for AstraZeneca, reflecting the frequency of China-specific analyst questions given China's 13–20% revenue share.

## 3.3 Score Distribution

**Table 4:** Distribution of Human Stance Scores ($n = 175$)

| Score | Label | Count | % |
|---|---|---|---|
| 1 | Committed Expansion | 35 | 20.0 |
| 2 | Optimistic Engagement | 103 | 58.9 |
| 3 | Cautious Localisation | 29 | 16.6 |
| 4 | Active Hedging | 7 | 4.0 |
| 5 | Strategic Retreat | 1 | 0.6 |

The concentration of scores in the 1-2 range is itself a substantive finding: even during a period of escalating US-China geopolitical tension, the dominant mode of executive communication remained positive or neutral toward China.[9]

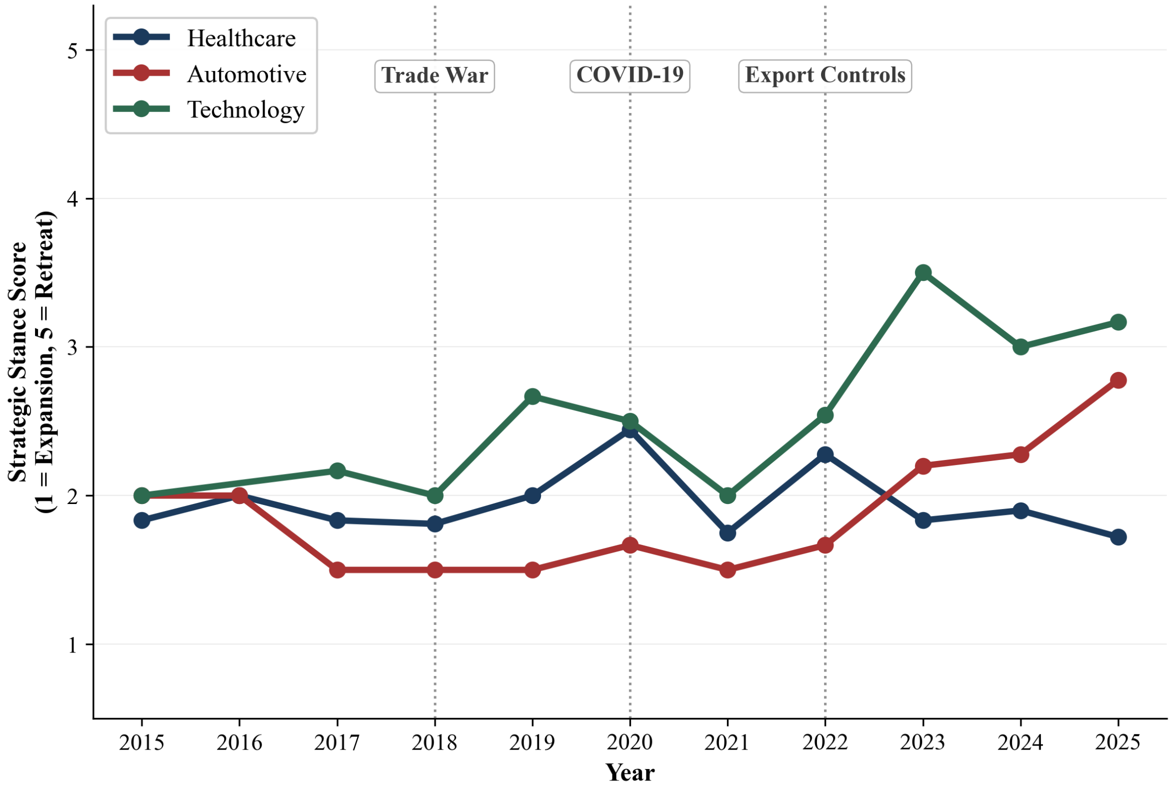
[9] This raises an important interpretive question about whether CEO communications lag operational realities. The gap between NVIDIA CEO Jensen Huang's compliance-focused messaging and CFO Colette Kress's financial disclosures ("material adverse impact," "$4.5 billion charge") suggests that CEO statements may systematically understate the severity of strategic repositioning.

# 4 Findings

## 4.1 Sector-Level Trajectories

Figure 1 presents sector-level mean stance scores over time.

**Figure 1: Sector-Level Mean CEO Strategic Stance Toward China, 2015–2025**



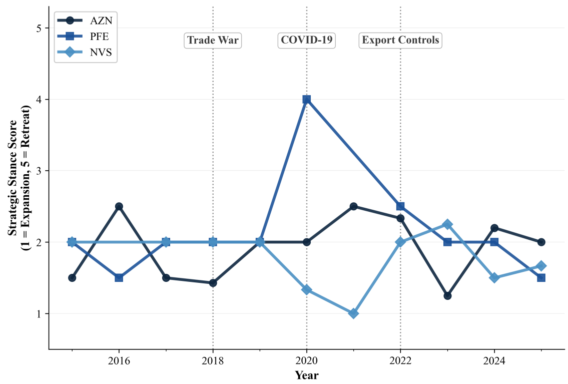

*Notes.* Each line represents the unweighted mean of firm-year stance scores within a sector. The y-axis measures strategic stance on a 1–5 ordinal scale (1 = Committed Expansion, 5 = Strategic Retreat). Vertical dotted lines mark the US-China trade war onset (2018), COVID-19 pandemic (2020), and US semiconductor export controls (2022). $N = 80$ firm-year observations across 9 firms.

Three distinct sector-level patterns are visible. **Healthcare** maintains the most stable and optimistic trajectory, oscillating between 1.5 and 2.5 throughout the period with no visible trend. **Automotive** shows a gradual upward drift post-2022, reflecting increasing competitive pressure from domestic Chinese EV manufacturers. **Technology** displays the most dramatic shift, with the sector mean rising sharply from approximately 2.0 in 2021 to above 3.0 by 2023, driven by NVIDIA's export-control-forced deterioration.

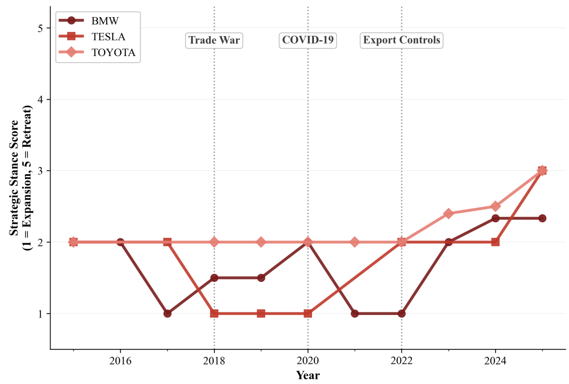
## 4.2 Firm-Level Trajectories

Figure 2 presents individual firm trajectories, grouped by sector, enabling within-sector comparison.

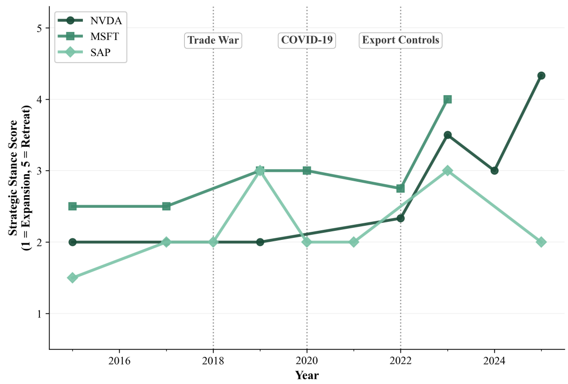
**Figure 2: Firm-Level CEO Strategic Stance Toward China, 2015–2025**




**Panel A: Healthcare**




**Panel B: Automotive**




**Panel C: Technology**

*Notes.* Lines represent annual mean stance score where multiple observations exist per firm-year. The y-axis measures strategic stance on a 1–5 ordinal scale (1 = Committed Expansion, 5 = Strategic Retreat). Vertical dotted lines mark key policy inflection points: US-China trade war (2018), COVID-19 (2020), and US semiconductor export controls (2022). $N = 175$ coded observations across 9 firms.

### 4.2.1 Healthcare: Consistent Optimism with Pricing Pressure

**AstraZeneca** (mean 1.95, $n = 44$) presents the richest longitudinal narrative. From 2015 to 2019, the firm exhibited strong Committed Expansion, anchored by CEO Pascal Soriot's personal engagement—including a dedicated China & Emerging Markets Investor Day in 2018 where EVP Leon Wang outlined a comprehensive strategy spanning 15,000+ nebulization centres and the Wuxi Innovation Centre. China revenue grew 25% in 2018 and 35% in 2019. The inflection came in 2021–2022, when Volume-Based Procurement (VBP) and COVID lockdowns created a "transition year" with mid-single-digit revenue decline. By 2023, Soriot declared "China is back" and announced a Qingdao manufacturing plant. In 2024 Q3, an investigation into company practices created temporary uncertainty,[10] but 2025 saw a new R&D centre in Beijing—indicating the trajectory re-

[10] AstraZeneca's Q3 2024 commentary explicitly addresses the investigation while reaffirming commitment: "We remain committed to our presence in China, and we will continue to invest."

turned to expansion.

**Novartis** (mean 1.75, $n = 24$) exhibits the lowest mean score of any firm in the dataset. CEO Vas Narasimhan's 2019 reframing of VBP as "freeing $30 billion for innovation" was a strategic landmark, positioning pricing reform as opportunity rather than threat. By 2021, the firm was treating China "the same as any other country" for portfolio planning. The 2024 announcement of a $100 million radioligand therapy manufacturing facility in China represents the strongest single committed expansion signal in the post-2022 dataset.The Novartis trajectory is distinctive because the October 2023 Sandoz spinoff removed the generics business most exposed to VBP pricing pressure, leaving the "pure innovative medicines" company with a structurally more positive China outlook.

**Pfizer** (mean 2.00, $n = 26$) follows a two-speed narrative. The established medicines business suffered a sharp VBP-driven decline: Upjohn China revenues fell 20% in Q2 2019 following the March 2019 VBP implementation for Lipitor and Norvasc. However, the innovative biopharmaceutical business grew 42% in Q3 2019, and CEO Albert Bourla simultaneously described China as home to a "very strong research organisation" with 19 new filings. By 2025, Bourla exhibited explicit dual framing: characterising China as "an opportunity" for Pfizer's business while warning Congress about China's biotech competitiveness.[11]

#### 4.2.2 Automotive: Localisation as Strategic Response

**BMW** (mean 1.80, $n = 20$) displays a peak-then-moderate trajectory. The 2017–2021 period represents sustained Committed Expansion: 595,000 vehicles sold in China (largest single market), the iX3 produced in Shenyang for global export, and the landmark 2021 acquisition of a 75% stake in BBA — making BMW the first foreign automaker to take majority control of a Chinese joint venture.[12] CEO Oliver Zipse met Premier Li Keqiang twice within his first three months in office (2019) and later accompanied Chancellor Scholz on an official China visit during Q3 2022. Post-2022, the language shifts to "cautiously optimistic" and "globally balanced footprint"—the emergence of hedging vocabulary alongside continued investment.

**Tesla** (mean 1.83, $n = 12$) follows a distinctive committed-expansion arc driven by the Shanghai Gigafactory. From the 2018 announcement (the first wholly-owned foreign auto factory in China, with no joint venture partner), through the 11-month construction and 2020 operational ramp-up with local supply chain reaching 70%+ localisation,

[11] Pfizer's trajectory highlights the risk of relying solely on a single composite score: the firm's stance toward China varies substantially depending on whether one examines the commercial business (optimistic), the R&D business (committed), or the geopolitical commentary (cautious).

[12] The BBA stake increase from 50% to 75% was enabled by the 2018 removal of foreign equity caps in the automotive sector. BMW's decision to exercise this option immediately, while other OEMs hesitated, represents a strong revealed-preference signal of China commitment.

Tesla exhibited peak Committed Expansion. Post-2022, CEO Elon Musk acknowledged Chinese competitors would "pretty much demolish most other car companies" if trade barriers were removed—the strongest competitive threat language in the dataset—but continued to position Tesla as uniquely capable of competing.[13] By 2025, a dual supply chain emerged (China LFP cells for rest-of-world markets, US manufacturing for domestic), representing the first Active Hedging signal from Tesla.

**Toyota** (mean 2.29, $n = 14$) shows the most pronounced within-sector shift. The FY2023 earnings call (May 2023) was the richest single transcript, with 41 China mentions. EVP Nakajima's admission that development in Japan "takes too long" and the need to accelerate local R&D represents a significant Cautious Localisation signal. By November 2024, CFO Miyazaki offered the dataset's most candid assessment: "we really endured, and we somehow managed to achieve about 90% of the previous year's sales"—explicitly acknowledging competitive pressure from domestic EV manufacturers while maintaining that Toyota's hybrid technology provides a structural advantage.

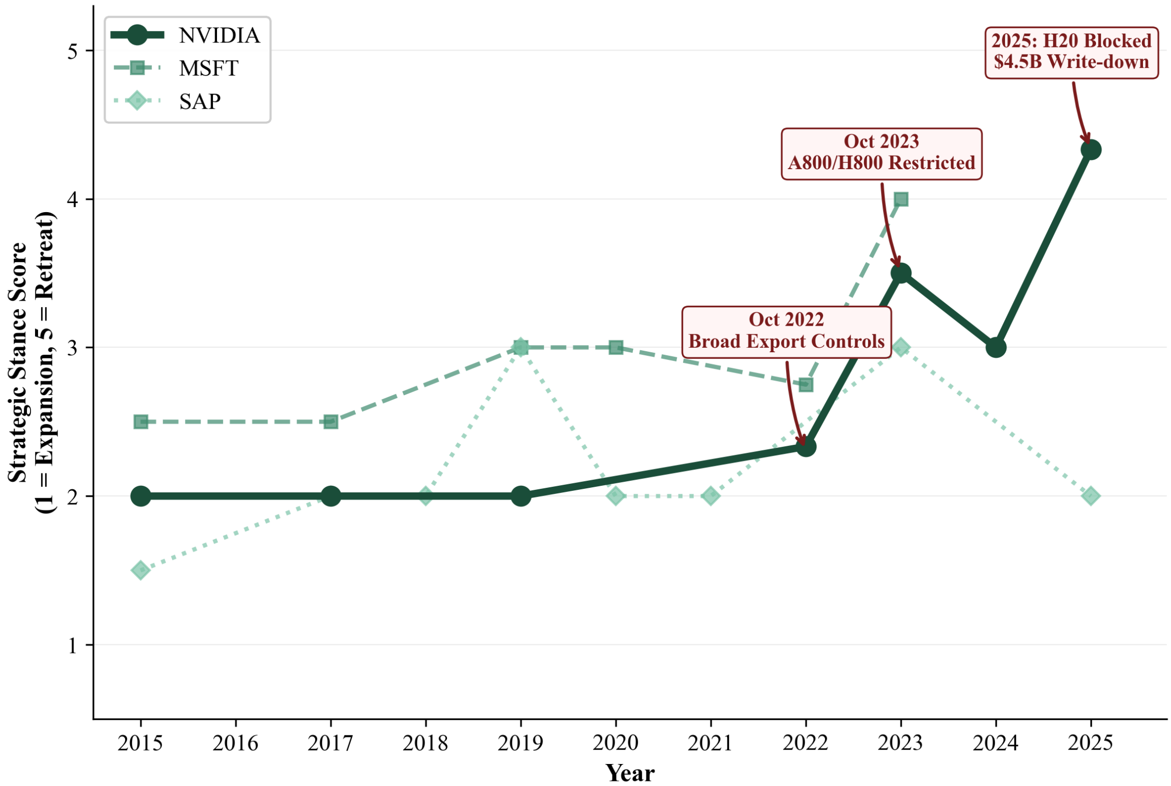
#### 4.2.3 Technology: The Bifurcation Story

**NVIDIA** (mean 3.00, $n = 12$) presents the most dramatic trajectory in the dataset. From 2015 to mid-2022, the firm maintained Score 2 (Optimistic Engagement), with China as the largest gaming market and an important data centre customer. The October 2022 export controls triggered an immediate shift to Score 3 (Cautious Localisation), as NVIDIA developed A800/H800 compliant alternatives. The October 2023 expansion of controls restricted even these alternatives, pushing the firm to Score 4 (Active Hedging)—CFO Kress disclosed that China represented 20–25% of data centre revenue and would "decline significantly." The 2025 H20 export ban resulted in the single Score 5 observation in the dataset: a $4.5 billion inventory charge, $8 billion single-quarter revenue loss, and a disclosure that China had fallen to "low single-digit percentage" of data centre revenue.[14]

[13] Tesla's wholly-owned factory structure is analytically significant: unlike BMW and Toyota, Tesla has no joint venture partner whose interests could constrain exit decisions, making its continued investment a purer signal of strategic commitment.

[14] The NVIDIA case is analytically important because it represents a *natural experiment* in policy-forced strategic retreat: each step in the stance trajectory can be traced to a specific, dated regulatory action, enabling clean identification of the causal mechanism.

**Figure 3: Technology Sector: Export Control Impact on NVIDIA vs Software Firms**




*Notes.* NVIDIA's stance trajectory (solid line) is plotted against Microsoft and SAP (dashed/dotted) to isolate the semiconductor-specific export control effect. Annotated events mark the October 2022 broad export controls, October 2023 A800/H800 restrictions, and 2025 H20 export ban with the associated $4.5 billion inventory charge. Score scale: 1 = Committed Expansion, 5 = Strategic Retreat.

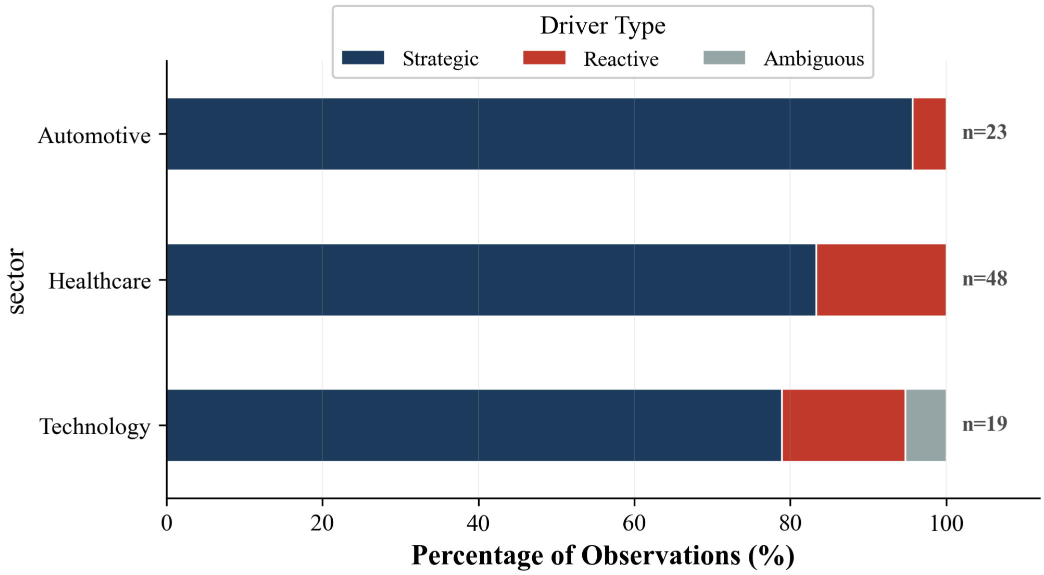
**Microsoft** (mean 2.75, $n = 12$) occupies a structurally distinct position. The 21Vianet partnership—making Azure "the only public cloud to operate legally in China"—represents a unique compliance architecture that insulates the firm from export control risk while enabling continued market participation. CEO Nadella's sovereignty framing ("data sovereignty and security is not going to go away ever") positions geopolitical fragmentation as a *market opportunity* rather than a risk.[15]

**SAP** (mean 2.09, $n = 11$) shows the clearest CEO-transition effect. Under Bill McDermott (2015–2019), China was framed as "deeply embedded" in SAP's strategy, with positive references to One Belt One Road. Under Christian Klein (2020–present), the language shifted to geographic diversification and data sovereignty, with CFO Asam characterising China as "largely still an on-prem market" in 2023.

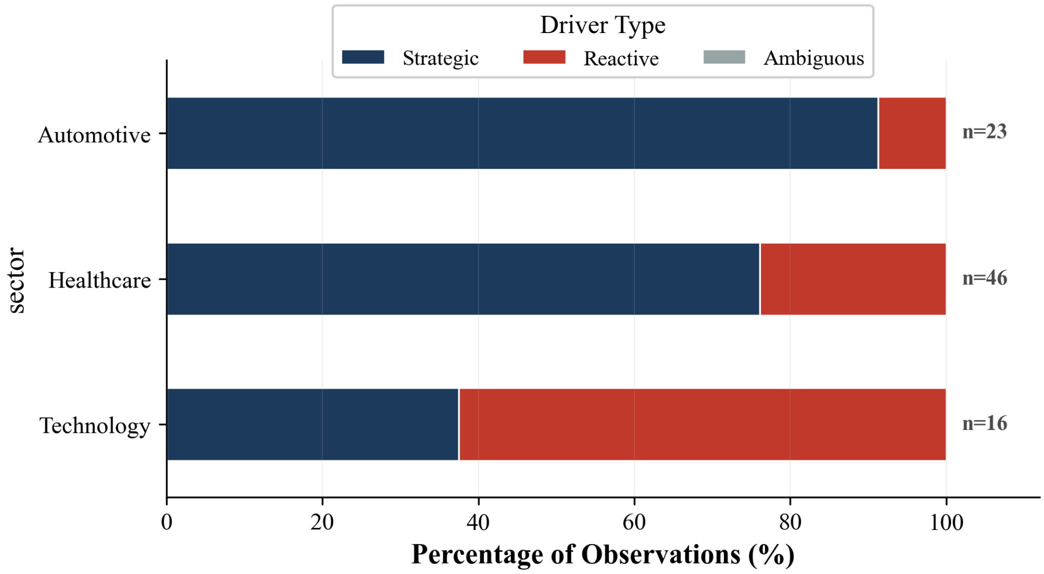
[15] Microsoft's 2023 characterisation of China as a "nation-state cyber adversary" alongside Russia and North Korea represents a qualitative shift not fully captured by the ordinal scale. The firm simultaneously maintains commercial engagement in China while framing China as a security threat—a dual stance that pushes the boundaries of the five-point classification.

## 4.3 Reactive vs Strategic Positioning

**Figure 4: Reactive vs Strategic Stance Drivers by Sector and Period**




**Panel A: Pre-2022**




**Panel B: Post-2022**

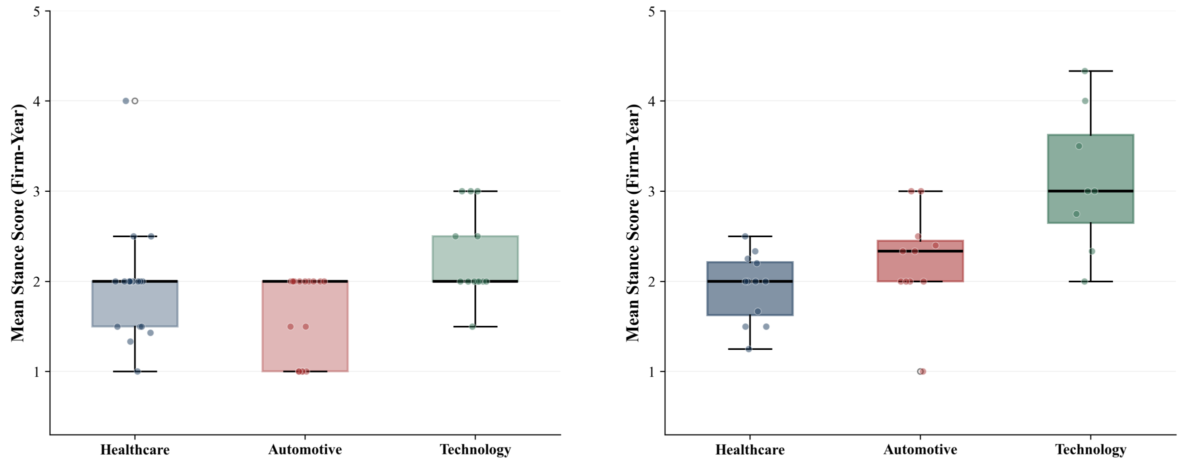
*Notes.* Each bar shows the percentage of coded observations classified as Strategic (proactive positioning), Reactive (policy-forced response), or Ambiguous. Pre-2022 captures the pre-export-control baseline; Post-2022 captures the period of US semiconductor export controls and intensified geopolitical competition. $n$ values denote total observations per sector in each period.

Of 175 observations, 139 (79%) were classified as Strategic and 35 (20%) as Reactive. The share of Reactive observations increases sharply for Technology post-2022 (from ∼10%

to ∼50%), driven by NVIDIA's export control responses.[16] Healthcare and Automotive remain predominantly Strategic in both periods.

## 4.4 Statistical Tests

**Figure 5: Stance Score Distribution by Sector and Period**




**Panel A: Pre-2022** **Panel B: Post-2022**

*Notes.* Box plots show the distribution of firm-year mean stance scores within each sector. The thick black line denotes the median; box edges represent the interquartile range (25th–75th percentile); whiskers extend to the most extreme observation within 1.5× IQR. Individual dots represent firm-year observations with slight horizontal jitter for visibility. Comparing Panel A to Panel B reveals the Technology sector's distributional shift upward post-2022.

Table 5 reports formal statistical tests on the firm-year panel.[17] The Welch's $t$-test confirms a statistically significant overall shift in mean stance score from pre-2022 (1.93) to post-2022 (2.35), with large effect sizes in Automotive ($d$ = 1.16) and Technology ($d$ = 1.28). The Mann-Kendall test identifies a significant monotonic increasing trend only for Technology ($Z = 2.24$, $p = 0.025$), confirming that the Technology sector is the only one experiencing a sustained directional shift rather than cyclical fluctuation. The Kruskal-Wallis test rejects the null hypothesis of equal distributions across sectors ($H = 13.87$, $p = 0.001$).

[16] The Reactive/Strategic distinction is analytically important for interpreting the information content of CEO communications. Reactive statements are responses to known policy constraints—their informational value lies in how firms describe adaptation. Strategic statements reflect genuine managerial choice—their informational value lies in revealed strategic priorities.

[17] All statistical results should be interpreted with appropriate caution given the small sample size ($n = 80$ firm-year observations). These tests describe patterns in the coded data; they are not causal estimates.

**Table 5:** Statistical Tests on Firm-Year Panel

| Test | Statistic | $p$-value | Interpretation |
|---|---|---|---|
| **Pre- vs Post-2022 structural break (Welch's $t$-test)** | | | |
| Overall | $t = -2.65$ | 0.011 | Significant |
| Healthcare | $t = 0.15$ | 0.878 | Not significant |
| Automotive | $t = -2.93$ | 0.009 | Significant ($d = 1.16$) |
| Technology | $t = -2.71$ | 0.021 | Significant ($d = 1.28$) |
| **Mann-Kendall monotonic trend test (sector-level)** | | | |
| Healthcare | $Z = -0.47$ | 0.640 | No trend |
| Automotive | $Z = 1.56$ | 0.120 | No trend |
| Technology | $Z = 2.24$ | 0.025 | Increasing trend |
| **Cross-sector comparison** | | | |
| Kruskal-Wallis | $H = 13.87$ | 0.001 | Significant differences |
| Mann-Whitney U (Tech vs Others) | $U = 159.5$ | 0.001 | Tech significantly higher |

*Notes*: Cohen's $d$ is reported for significant sector-level results. Mann-Whitney U compares Technology vs non-Technology firms in the post-2022 period. All tests conducted on the 80-observation firm-year panel.

# 5 Discussion

## 5.1 Policy Exposure as the Organising Variable

The central finding of this analysis is that strategic stance toward China is best understood through the lens of *policy exposure type* rather than sector membership. Three distinct regulatory regimes produce three distinct executive response patterns:

- **US extraterritorial controls** (affecting NVIDIA) produce forced, reactive, stepwise deterioration in stance. Each regulatory action is followed by a measurable downward shift, with limited managerial discretion over timing or magnitude.
- **Host-country market reforms** (affecting all three healthcare firms via VBP) produce temporary dips followed by recovery, as firms adapt product portfolios and pricing strategies. The key distinction is that VBP constrains margins but does not restrict market access—executives retain strategic agency.
- **Competitive pressure from domestic firms** (affecting all three automotive firms) produces gradual localisation without retreat. Executives frame adaptation as offensive ("model offensive," "local for local") rather than defensive.

## 5.2 The "Local for Local" Ambiguity

German automotive OEMs and pharmaceutical firms employ localisation language that simultaneously signals market commitment to host-country audiences and risk mitigation to investors. BMW's "globally balanced footprint" framing and AstraZeneca's "serving the Chinese market from China" rhetoric serve dual functions that cannot be disentangled from CEO communications alone.[18] Triangulating CEO statements with operational data—CapEx flows, patent filings by jurisdiction, supplier network shifts—would be necessary to distinguish strategic intent from rhetorical framing.

## 5.3 CEO Communications as Imperfect Signals

The gap between NVIDIA CEO Jensen Huang's compliance-focused messaging ("right products for the Chinese market") and CFO Colette Kress's financial disclosures ("material adverse impact," "$4.5 billion charge") suggests that C-suite communications may lag and understate operational realities. This is consistent with voluntary disclosure theory: under regulatory uncertainty, executives have incentives to signal stability and compliance while financial disclosures reveal the operational impact with a lag.[19]

## 5.4 Limitations

Several limitations should be acknowledged. First, 175 observations across 9 firms provide limited statistical power; all results should be interpreted as descriptive patterns rather than causal estimates. Second, selection into public commentary is non-random—firms that discuss China more frequently produce more codable observations, and this propensity to discuss China may itself correlate with China exposure, creating potential bias. Third, the ordinal scale compresses rich qualitative variation into five categories; Pfizer's simultaneous optimism about commercial opportunities and concern about competitive threats from Chinese biotech, for instance, maps uncomfortably onto a single score. Fourth, the study captures *expressed* sentiment, not necessarily *revealed* strategic behaviour — a gap that is likely widest for firms under the most intense geopolitical scrutiny.

[18] This dual-use quality of localisation rhetoric has implications for research design: studies that code "localisation" as either commitment or hedging (but not both) may systematically misclassify a substantial share of executive statements.

[19] This CEO-CFO divergence pattern, where the CEO provides qualitative reassurance and the CFO quantifies the damage appeared consistently in NVIDIA transcripts across three successive rounds of export controls. Future research could systematically code CEO vs CFO language within the same firm to measure this divergence.

# 6 Conclusion

This report constructs a longitudinal dataset of CEO strategic stance toward China across nine multinational firms (2015–2025), demonstrating that LLM-assisted content analysis achieves high intercoder reliability ($\alpha = 0.931$) for ordinal classification of strategic executive communications. The key findings are:

1. **Dominant Optimism:** The overall mean stance score of 2.09 indicates that, across the decade, multinational executives maintained a predominantly positive orientation toward China, even as geopolitical tensions escalated.

2. **Sectoral Divergence:** Statistically significant structural breaks emerge post-2022 for Automotive ($p = 0.009$) and Technology ($p = 0.021$), but not for Healthcare ($p = 0.878$).

3. **The Technology Deterioration:** Technology is the only sector with a significant monotonic trend toward more cautious positioning ($Z = 2.24$, $p = 0.025$), driven by NVIDIA's export-control-forced trajectory.

4. **Information Value:** The distinction between Reactive and Strategic positioning illuminates differential informational content: reactive statements reveal compliance constraints, while strategic statements reveal genuine managerial priorities.

These findings suggest that China's risk assessment, whether by corporate strategists, policymakers, or academic researchers — must be disaggregated by the specific regulatory mechanism at play, rather than relying on broad sector-level or country-level generalisations.

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